Supplementary Material

Affordable Real Style Transfer: Training Wavelet-Decoupled Rectified Flow on a Single RTX 3060 in Minutes

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1 KL Divergence Verification of Gaussian Assumption

We verify the Gaussian approximation for WCT by computing the KL divergence between the empirical distribution of each wavelet subband and a fitted multivariate Gaussian, on the Distinct5-WikiArt test set (750 latent tensors).

For each subband $h_i \in \mathbb{R}^{4\times16\times16}$, we flatten the spatial dimensions to obtain feature vectors of dimension $d = 4$ (the channel dimension). We then fit a d-dimensional Gaussian $\mathcal{N}(\hat{\mu}_i, \hat{\Sigma}_i)$ and compute the KL divergence

$$D_{\mathrm{KL}}(p_{\mathrm{emp}} \| \mathcal{N}(\hat{\mu}_i, \hat{\Sigma}_i)) = \mathbb{E}_{p_{\mathrm{emp}}}[\log p_{\mathrm{emp}} - \log \mathcal{N}(x; \hat{\mu}_i, \hat{\Sigma}_i)]. \tag{1}$$

The results are:

Subband	LL	LH	HL	HH
KL divergence (nats)	0.031	0.042	0.038	0.047
% of samples within 2σ	96.2	94.8	95.3	94.1

All subbands have KL divergence below 0.05 nats, and over 94% of samples fall within the 2σ ellipse of the fitted Gaussian. This confirms that the VAE KL regularization successfully Gaussianizes the latent space, making second-order moment matching (WCT) well-justified.

2 Proof of Theorem 1 (Local Collapse in Euclidean Latents)

We restate Theorem 1 in the main text and provide a self-contained proof.

Theorem 1 (Local Collapse). *Let $\theta_0 \in \mathcal{M}_0$ be stationary in the normal directions:*

$$\nabla_\perp \mathcal{L}_{\mathrm{FM}}(\theta_0) = \nabla_\perp \mathcal{L}_{\mathrm{sty}}(\theta_0) = 0. \tag{2}$$

Let L_f^{low} and L_f^{high} be the Lipschitz constants of $\nabla \mathcal{L}_{\mathrm{FM}}$ restricted to the low- and high-frequency latent subspaces, and let L_s be the Lipschitz constant of $\nabla \mathcal{L}_{\mathrm{sty}}$. If $w_f L_f^{\mathrm{high}} > w_s L_s$, then θ_0 is a strict local minimum of $\mathcal{L}_{\mathrm{euc}} = w_f \mathcal{L}_{\mathrm{FM}} + w_s \mathcal{L}_{\mathrm{sty}}$ on the normal space $T_{\theta_0}^\perp \mathcal{M}_0$.

Proof. Consider a normal perturbation $\delta \in T_{\theta_0}^\perp \mathcal{M}_0$ with $\|\delta\|$ sufficiently small. The stationarity condition implies $\nabla \mathcal{L}_{\mathrm{euc}}(\theta_0) = 0$. A Taylor expansion to second order yields

$$\mathcal{L}_{\mathrm{euc}}(\theta_0 + \delta) = \mathcal{L}_{\mathrm{euc}}(\theta_0) + \frac{1}{2}\delta^\top \nabla_\perp^2 \mathcal{L}_{\mathrm{euc}}(\theta_0)\delta + o(\|\delta\|^2). \tag{3}$$

The normal Hessian decomposes by the Lipschitz bounds:

$$\delta^\top \nabla_\perp^2 \mathcal{L}_{\mathrm{euc}}(\theta_0)\delta \geq w_f L_f^{\mathrm{high}} \|\delta\|^2 - w_s L_s \|\delta\|^2. \tag{4}$$

Here the first term is the lower bound from the flow-matching Hessian restricted to the high-frequency subspace (where style-conditioned components live), and the second term is the upper bound from the style loss. The $1/f$ spectral structure of VAE latents implies $L_f^{\mathrm{low}} \gg L_f^{\mathrm{high}}$, but the collapse condition only requires the inequality in the high-frequency subspace.

By hypothesis, $w_f L_f^{\mathrm{high}} > w_s L_s$, so the quadratic form is strictly positive definite. Hence $\mathcal{L}_{\mathrm{euc}}(\theta_0 + \delta) > \mathcal{L}_{\mathrm{euc}}(\theta_0)$ for any sufficiently small $\delta \neq 0$, confirming θ_0 as a strict local minimum.

Remark. The condition $w_f L_f^{\mathrm{high}} > w_s L_s$ is satisfied by standard CNN backbones because: (1) the $1/f$ power spectrum of natural images makes L_f^{high} non-negligible even though L_f^{low} dominates; (2) the style-matching term $\mathcal{L}_{\mathrm{sty}}$ is bounded by the CLIP embedding space, which has a fixed Lipschitz constant; (3) the flow-matching weight w_f is typically larger than w_s in standard training configurations.

3 Proof of Proposition 1 (Exponential Content Decay)

We restate Proposition 1 in the main text.

Proposition 1 (Exponential Content Decay). *Let a style injection of strength $\alpha \in (0, 1)$ be applied at every solver step for n steps. The residual content fraction after n steps is*

$$r_n(\alpha) = (1 - \alpha)^n. \tag{5}$$

For $n = 8$, $r_8(0.5) = 3.9 \times 10^{-3}$. Endpoint-only injection ($n = 1$) yields $r_1(\alpha) = 1 - \alpha$, which is affine and preserves blend identifiability.

Proof. At each step, the injection operator I_α blends the current latent z_t toward the style reference by a factor α:

$$z_t \leftarrow (1-\alpha)z_t + \alpha \cdot \text{style\_signal}. \tag{6}$$

The content component of z_t is multiplied by $1-\alpha$ at each step. After n independent injections, the residual content fraction is the product of n such factors:

$$r_n(\alpha) = \prod_{k=1}^{n}(1-\alpha) = (1-\alpha)^n. \tag{7}$$

For $n = 1$, the residual is $1-\alpha$, which is an affine function of α. For $n \gg 1$, the decay is exponential, compressing the operational blending space and making the style-strength/content-preservation tradeoff non-identifiable.

4 Subband Energy Distribution on Distinct5-WikiArt

We measured the per-subband energy distribution on the Distinct5-WikiArt test set (750 latent tensors, each $4 \times 32 \times 32$).

	LL	LH	HL	HH
Mean energy fraction (%)	91.7	3.2	3.0	2.1
Std of energy fraction (%)	2.1	0.8	0.7	0.6
Style-induced relative change (%)	4.3	11.8	10.2	15.6

The LL subband carries 91.7% of the total Frobenius norm on average, confirming the $1/f$ spectral structure. The style-induced relative change (computed as the ratio of per-subband energy difference between content and target-style latents to the content energy) is 2–4$\times$ larger in LH/HL/HH than in LL, confirming that style differences are concentrated in the high-frequency subbands.